

Customer Churning Prediction Using LLM

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Introduction

Modern customer support requires efficient analysis of customer queries and automated response systems. This project combines traditional NLP with LLMs to create an enhanced support system.”

Background

Customer churning is when customers defect or stop using a company. This is problematic for companies, who seek to minimize this.

In pursuit of this goal, companies have employed various machine learning projects to predict the churning of a customer

- This project uses a pretrained BERT model
 - An LLM which excels at understanding words in context
- Used for prediction purposes in this project

Conclusion

BERT models on their own struggle with CHURN prediction without training of data

Numeric data seems to be a weak point, excels when given strings of words

Sentiment is mapped well with BERT/LLM

Methodology

Customer Churning Prediction

- Customer data was transformed into sentences with MASKS that predicted churning.
- Sentence was crafted from the perspective of the user
- BERT requires prompt to be “fill in the blank”
- Y/N prompts don't seem to work and causes model to output inappropriate results
- Filter words
 - Not churned:
 - Keep
 - Churned:
 - Cancel, abandon, terminate, end
- Wrote program to see what words appeared for filter
 - What to do with neutral/ambiguous words
 - Sell, enquire, ask, etc.
- Final prompt:
 - "I am undecided as to whether I should keep or cancel my internet service. However, I have decided and knowing all of what I have said before, I decided to [MASK] my internet service."

Customer Profile Analysis

- Generated a descriptive profile analysis from the data based off of factors like age gender marital status and dependents.
- The descriptions of the customers profile were generated using natural language so that they are clearer and follow this structure “Female adult customer, married and with dependents,...”

Customer Churn Explanation

- Generates explanations for why a customer might or might not churn in normal language based off of the data
- It used key factors like contract type, tenure length, tech support availability, and more to determine why the customer might have churned.

Customer Sentiment

- Validation Approach:**
 - Behavioral narratives for a balanced random sample of 150 customers using API calls (e.g. Gemini, OpenAi)
- API Availability (primary bottleneck)**
 - Processing speeds dominated by external API network latency, server load, account limitation
 - Performance Variabilities**
 - Off-Peak: 32 minutes, near-zero retries
 - Peak: 45 minutes, frequent retries due to higher traffic and error calls
- Secondary Constraints:**
 - Account Limits:** “Free Tier” with low RRM, forcing slow serial processing instead of batch/parallel processing
 - Hardware & Prompt Types:** Complex Chain-of-Thought (COT) vs A-Few Shot Prompting; CPU-based hardware took ~ 90 minutes for 150 COT

```
customer = 1
f"\"a simple billing question that was resolved quickly\",
f\"I have been a loyal customer for many years and I am very satisfied with the service.\",
f\"feeling valued as a loyal, long-term customer (TotalCharges: ${customer_row['TotalCharges']})\"

chosen_reason = f\"praising {random.choice(reasons)}\"

prompt = f\"\"\"
You are a senior customer behavior analyst. Based on {chosen_reason}, generate a behavioral narrative
You are a senior customer behavior analyst. Based on {chosen_reason}, generate a behavioral narrative

INSTRUCTIONS:
Infer the customer's likely habits, personality, and relationship with their services. Focus on:
1. **Household & Social Context:** (Analyze SeniorCitizen, Partner, Dependents)
2. **Tech-Savviness & Needs:** (Analyze InternetService, OnlineBackup, TechSupport, MultipleLines)
3. **Financial & Contractual Stance:** (Analyze Contract_Encoded, MonthlyCharges, and TotalCharges. No
4. **Overall "person" created by this data.**
\"\"\"
```

Few-shot Prompt Examples

Example 1 (Churned - Value Frustration): A moderate-use customer with DSL and no tech support. Narrative Style: Highlights frustration with perceived low value despite a one-year contract; leading to churn based on cost-effectiveness.

Example 2 (Retained - Loyal Family): A high-demand household with Fiber Optic and tech support on a two-year contract. Narrative Style: Focuses on deep loyalty, low financial risk, and satisfaction with a comprehensive service package.

```
data_set["SeniorCitizen"] = data_set["SeniorCitizen"].map({1: "a senior citizen", 0: "not a senior citizen"})

data_set["Partner"] = data_set["Partner"].map({"Yes": "a partner", "No": "no partner"})

data_set["Dependents"] = data_set["Dependents"].map({"Yes": "dependents", "No": "no dependents"})

data_set["PhoneService"] = data_set["PhoneService"].map({"Yes": "have phone service", "No": "do not have phone service"})
```

Results

Customer Churning Prediction

Model #1, DS #1

Model #1, DS #2

Model #1, DS #3

Model #2, DS #1

Metrics from left to right, top to bottom

- A: 0.491, 0.598, 0.565, 0.735
- P: 0.292, 0.260, 0.074, 0
- R: 0.647, 0.278, 0.055, 0

Base positive occurrence: 0.265

Customer Sentiment

Model #1, DS #2

Model #2, DS #2

Customer Journey Risk Map

Contract Type

Month-to-Month (Low Commitment)

Long-Term Contract (High Commitment)

Manual Payment (High Friction)

Automatic Payment (Low Friction)

CRITICAL RISK ZONE (High Friction, Low Commitment)

MODERATE RISK (Low Friction, Low Commitment)

MODERATE RISK (High Friction, High Commitment)

SAFE ZONE (Low Friction, High Commitment)